



From IoT to Mobile

Oniro's Technical Journey Toward Digital Independence

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The Digital Sovereignty Problem

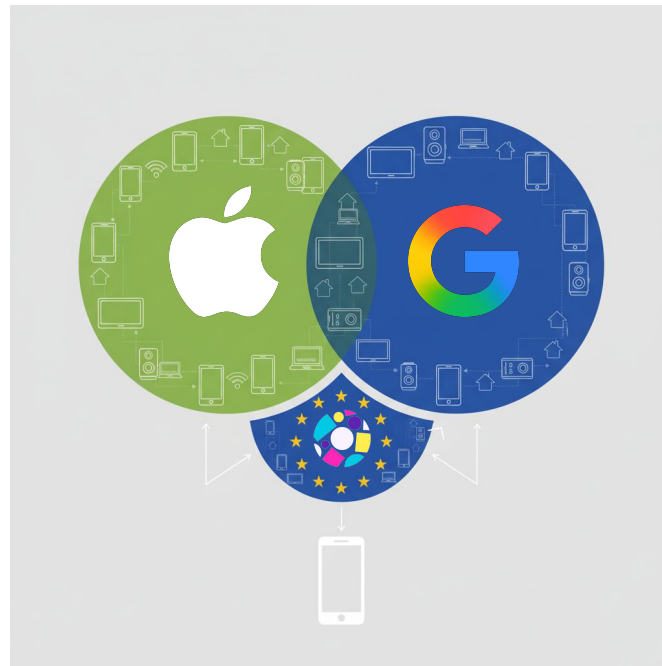


Two platforms dominate mobile computing

- Concentrated power & structural dependency
- Limited strategic options
- Privacy and security challenges

Europe needs alternatives that deliver:

- Strategic independence
- Multi-device interoperability
- Privacy compliance & local accountability



What is Oniro?

Open-source OS platform managed by Eclipse Foundation

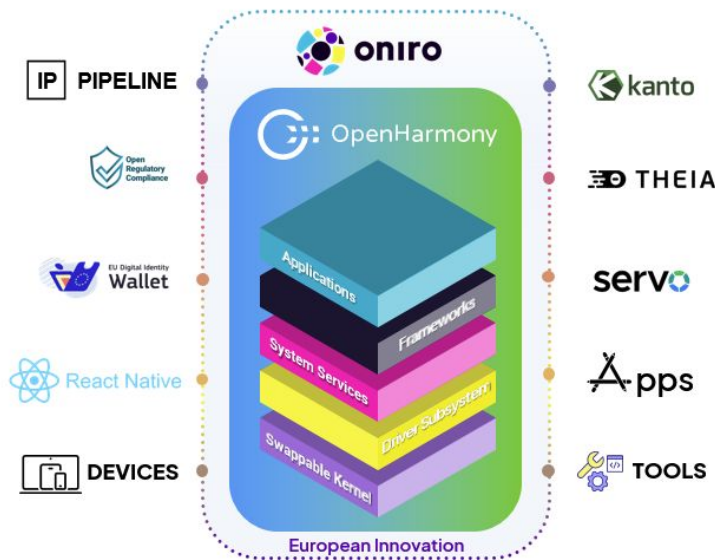
Built on OpenHarmony (a mature open-source project):

- Hundreds of products in production
- Thousands of contributors
- Millions of lines of code

Unique dual-foundation model:

- OpenAtom Foundation (China) + Eclipse Foundation (Europe)
- Shared technology, European governance
- Global scale with regional trust

Mission: Make cross-device platforms accessible for global adoption



From Foundations to Global Access



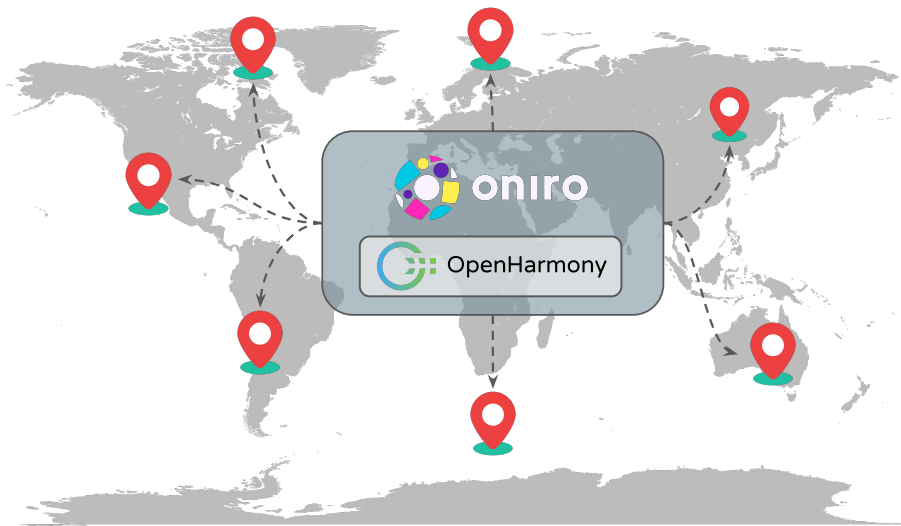
Why Oniro bridges OpenHarmony to the world:

Barriers outside China:

- Platform unfamiliarity (GitCode vs GitHub)
- Language barriers
- Limited access to familiar devices/tools

Oniro removes friction:

- GitHub mirrors with daily synchronization
- English-first documentation
- Globally available hardware support
- Developer-friendly tooling



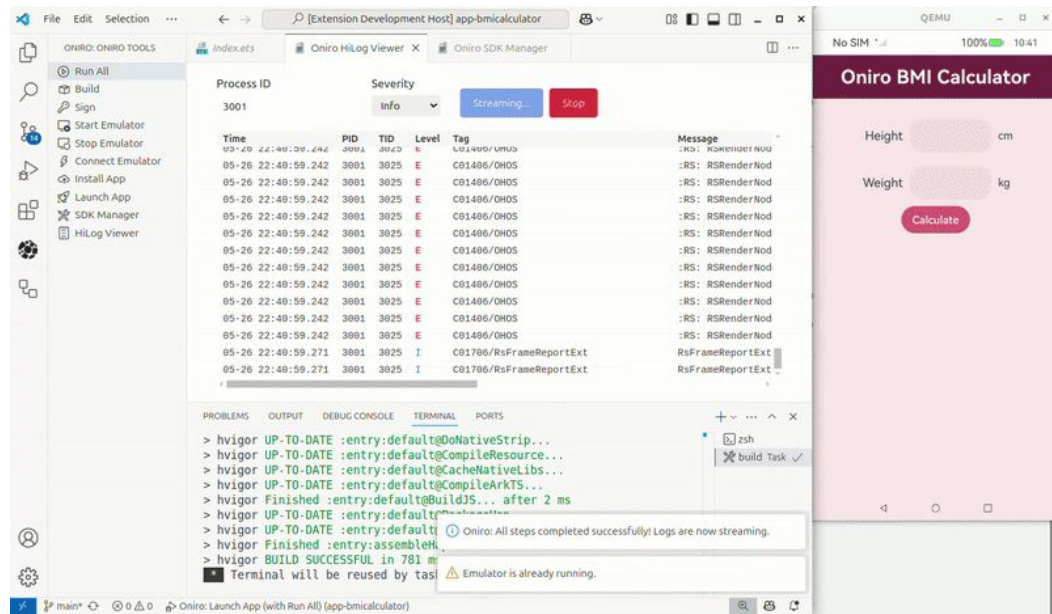
Developer Experience – Oniro IDE

Developer Experience – Oniro IDE



Impact:

- Fast development workflow in familiar editor
- Linux support (no Windows requirement)
- AI-assisted development ready
- Lowers entry barrier for new contributors



Testing Without Hardware – QEMU Emulator

QEMU-based Emulator

Technical approach:

- Started with Yocto meta-layer + LXC container
- Full graphical support (Mesa drivers)
- Native OpenHarmony integration
- x86_64 architecture → hypervisor acceleration

Enables:

- Instant testing + IDE integration
- CI/CD pipelines
- Rapid prototyping

OpenHarmony accessible to everyone—no physical device needed



Real Hardware – IoT to Mobile



Platform maturity across device types

IoT & Embedded:

- OpenHarmony reference boards (Hoperun)
- Raspberry Pi support (globally accessible)



Mobile devices:

- OpenHarmony Developer Phone (Unisoc chipset)
- OnePlus 6T (community port)
- European mobile reference device (proof-of-concept)



Goal: Expand to globally accessible hardware

Challenges of Mobile Hardware Porting



The proprietary driver problem strengthens the duopoly:

- Manufacturers don't open-source mobile drivers
- Drivers built for Android (Bionic libc), incompatible with alternatives
- Binary blobs create vendor lock-in and reinforce platform monopolies

Recent recognition: FSF's Librephone Initiative (October 2025)

- Aims to reverse-engineer obstacles to mobile phone freedom
- Close gaps between Android distributions and software freedom
- Led by Rob Savoye, building on LineageOS work
- "Making fully free software for modern phones will not be quick, easy, or cheap"



Challenges of Mobile Hardware Porting



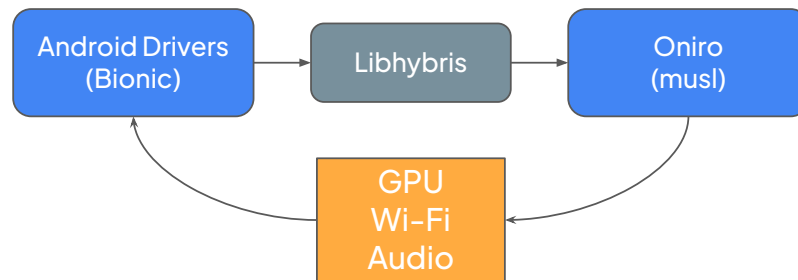
Oniro's pragmatic approach: LibHybris integration

- Adaptation layer translating Android calls to OpenHarmony's musl libc
- Enables GPU, Wi-Fi, peripheral support
- **Workaround, not ideal:** Binary blobs remain a black box
- First integration complete, more work ongoing

Bridging proprietary Android drivers for open systems



Libhybris



This could unlock Oniro for existing global mobile phones

Unified Cross-Device Architecture



One platform, multiple device classes:

System	Devices	Capabilities
Mini	Sensors, wearables	MCU, lightweight
Small	Cameras, routers	Security, media
Standard	Phones, tablets	Full UI, 3D GPU

Mini



Small



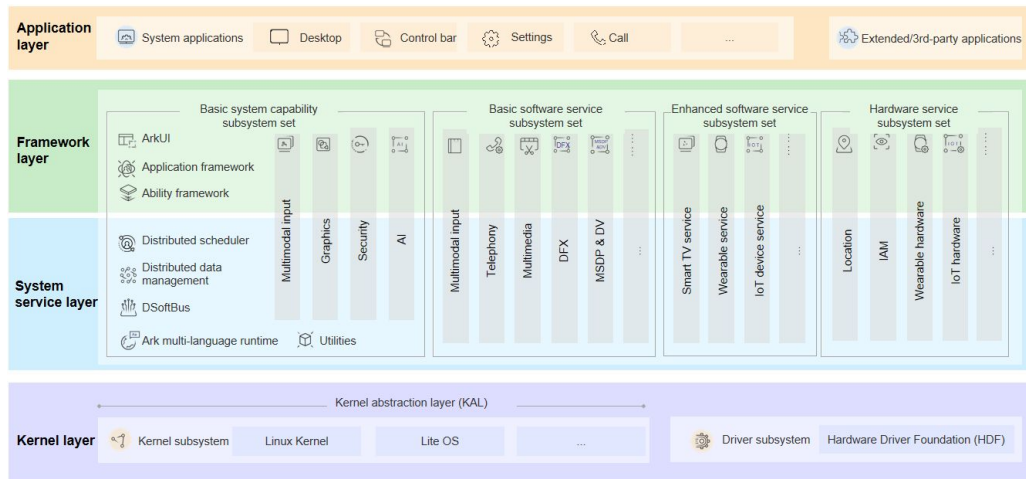
Standard



Unified Cross-Device Architecture

Key technologies enabling interoperability:

- Multi-kernel support: **LiteOS** for mini system, **Linux** for standard system
- HDF: Unified driver abstraction for portability
- DSoftBus: Seamless device discovery & communication
- Distributed capabilities: Data sync, remote app invocation, device virtualization ("Super Device")



Growing the App Ecosystem



Multiple strategies for app availability:

Infrastructure:

- Oniro App Store (F-Droid-like, open-source distribution)
- CI/CD for apps and system images
- Community apps: SoundCloud, Telegram...

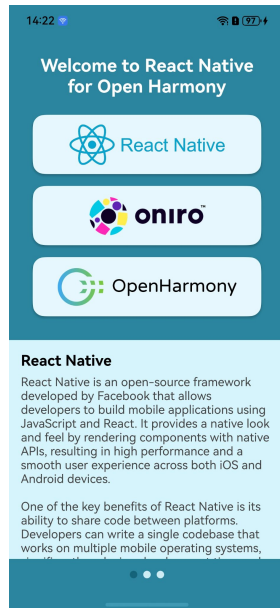


Growing the App Ecosystem

Cross-platform frameworks:

- React Native (hundreds of apps already available)
- Kotlin Multiplatform & Flutter (under research)
- Rust frameworks (Tauri, Robius, Dioxus)

Targeting multiple platforms with a single codebase.

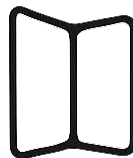
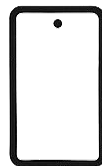
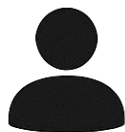
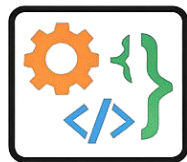


Roadmap



Future works:

- Richer app ecosystem through
- Wider third party frameworks integration
- Stronger global developer ecosystem
- More phones & form factors



Oniro Community



- Connect with other developers and the Oniro team.
- Ask questions, seek help, and engage in discussions.
- Join the [Oniro Matrix channel](#)

 Element | Oniro Project



Oniro Project

M

Mats Lundgren

23:30

How is our 5.0 Rebase going?

The HOS Next Public Beta and matching IDE build was released earlier this week.

<https://developer.huawei.com/consumer/cn/download/>

HOS next public beta build is 5.0.3.900

So now there are millions of devices that can get the HOS Next 5.0.3.900 build which is based on OH 5.0.

23:32

It also means it should be easy to write HOS Next apps that can run on Oniro OS based on OH 5.0
With the caveat on potential HOS Next SDK dependencies.

[matrix]



Human zone!
AI not invited 😊

Join Us in Shaping the Future



Contribute to Oniro development.
Let's create an open, secure mobile
ecosystem together.

Think Global and Code Local



oniroproject.org



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